INCH-POUND
MIL-DTL-27426/3B
w/AMENDMENT 1
10 May 2022
SUPERSEDING
MIL-DTL-27426/3B
1 March 2022

DETAIL SPECIFICATION SHEET

RINGS, RETAINING, SPIRAL (UNIFORM CROSS SECTION) MEDIUM DUTY INTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27426.

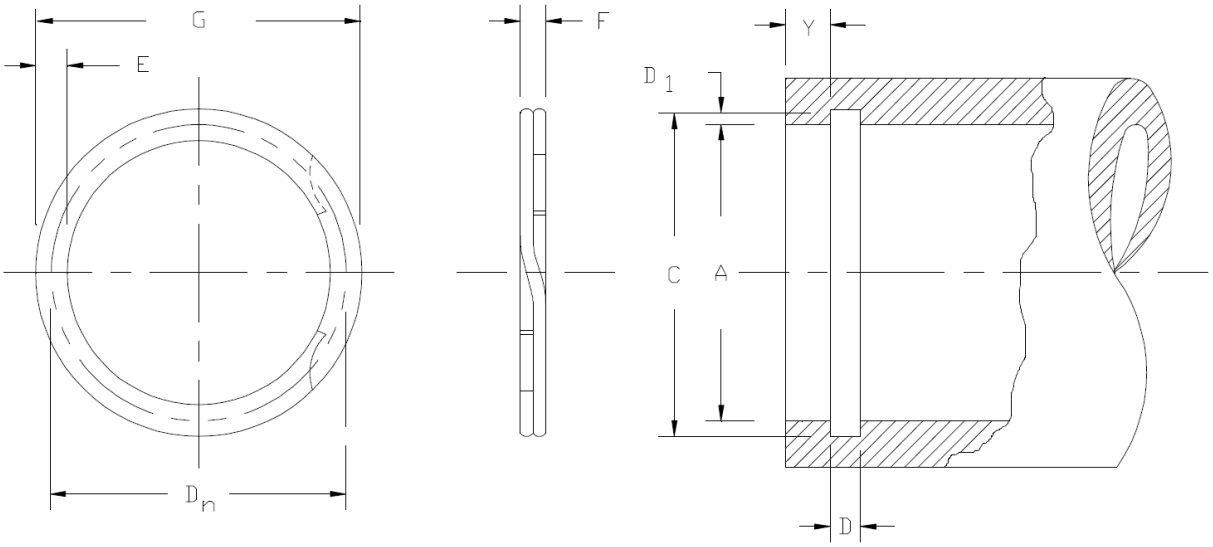


FIGURE 1. External ring.

Maximum bottom groove radius is 0.005 inches for rings up to 0.946 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency - Aviation, ATTN: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TABLE I. Medium duty internal retaining rings.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH D_1
	A		G	Tol*	F	Tol	E	Tol	C	Tol	D	Tol	
0100	0.500	12.7	0.532		0.025		0.045		0.526		0.030		0.013
0101	0.512	13.0	0.544		0.025		0.045		0.538		0.030		0.013
0102	0.531	13.5	0.564		0.025		0.045		0.557		0.030		0.013
0103	0.562	14.3	0.594		0.025		0.045		0.588		0.030		0.013
0104	0.594	15.1	0.626		0.025		0.045		0.619		0.030		0.013
0105	0.625	15.9	0.658		0.025		0.045		0.651	±0.002	0.030		0.013
0106	0.656	16.7	0.689		0.025		0.045		0.682		0.030		0.013
0107	0.687	17.5	0.720		0.025		0.045		0.713		0.030		0.013
0108	0.718	18.3	0.751		0.025		0.045		0.744		0.030		0.013
0109	0.750	19.0	0.790		0.031		0.065		0.782		0.036		0.016
0110	0.777	19.7	0.817		0.031		0.065		0.808		0.036		0.016
0111	0.781	19.8	0.821	+0.013 -0.000	0.031		0.065		0.812		0.036		0.016
0112	0.812	20.6	0.853		0.031		0.065		0.843		0.036		0.016
0113	0.843	21.4	0.889		0.031		0.065		0.880		0.036	+0.003 -0.000	0.019
0114	0.866	22.0	0.913		0.031		0.065		0.903		0.036		0.019
0115	0.875	22.2	0.922		0.031		0.065		0.912		0.036		0.019
0116	0.906	23.0	0.949		0.031		0.065		0.939	±0.003	0.036		0.019
0117	0.938	23.8	0.986		0.031		0.065		0.975		0.036		0.019
0118	0.968	24.4	1.025		0.037		0.075		1.015		0.042		0.021
0119	0.987	25.0	1.041		0.037	±0.002	0.075		1.030		0.042		0.021
0120	1.000	25.4	1.054		0.037		0.075		1.043		0.042		0.021
0121	1.023	26.0	1.078		0.037		0.075		1.066		0.042		0.021
0122	1.031	26.3	1.084		0.037		0.075		1.074		0.042		0.021
0123	1.062	27.0	1.117		0.037		0.075		1.104		0.042		0.021
0124	1.093	27.8	1.147		0.037		0.075		1.135		0.042		0.021
0125	1.125	28.6	1.180		0.037		0.075		1.167		0.042		0.021
0126	1.156	29.3	1.210		0.037		0.075		1.198		0.042		0.021
0127	1.188	30.2	1.249		0.043		0.085	+0.003 -0.005	1.236		0.048		0.024
0128	1.218	31.0	1.278		0.043		0.085		1.266		0.048		0.024
0129	1.250	31.7	1.312		0.043		0.085		1.298		0.048		0.024
0130	1.281	32.6	1.342	+0.015 -0.000	0.043		0.085		1.329		0.048		0.024
0131	1.312	33.3	1.374		0.043		0.085		1.360	±0.004	0.048		0.024
0132	1.343	34.1	1.408		0.043		0.085		1.395		0.048		0.026
0133	1.375	34.9	1.442		0.043		0.095		1.427		0.048		0.026
0134	1.406	35.8	1.472		0.043		0.095		1.458		0.048		0.026
0135	1.437	36.5	1.504		0.043		0.095		1.489		0.048		0.026
0136	1.456	37.0	1.523		0.043		0.095		1.508		0.048		0.026
0137	1.468	37.3	1.535		0.043		0.095		1.520		0.048		0.026
0138	1.500	38.1	1.567		0.043		0.095		1.552		0.048		0.026
0139	1.562	39.6	1.634		0.049		0.108		1.617		0.056		0.028
0140	1.574	40.0	1.649		0.049		0.108		1.633		0.056		0.030
0141	1.625	41.7	1.701		0.049		0.108		1.684		0.056	+0.004 -0.000	0.030
0142	1.653	42.0	1.730		0.049		0.108		1.712		0.056		0.030
0143	1.687	42.8	1.768		0.049		0.118		1.750		0.056		0.031
0144	1.750	44.4	1.834	+0.020 -0.000	0.049		0.118		1.813		0.056		0.031
0145	1.813	46.0	1.894		0.049		0.118		1.875	±0.005	0.056		0.031
0146	1.850	47.0	1.937		0.049		0.118		1.917		0.056		0.034
0147	1.875	47.6	1.960		0.049	±0.003	0.118		1.942		0.056		0.034
0148	1.938	49.2	2.025		0.049		0.118		2.005		0.056		0.034
0149	2.000	50.8	2.091		0.049		0.128		2.071		0.056		0.035
0150	2.047	52.0	2.138		0.049		0.128		2.118		0.056		0.035
0151	2.062	52.3	2.154		0.049		0.128		2.132		0.056		0.035
0152	2.125	53.9	2.217	+0.025 -0.000	0.049		0.128		2.195		0.056		0.035
0153	2.165	55.0	2.260		0.049		0.138		2.239	±0.006	0.056		0.037
0154	2.188	55.5	2.284		0.049		0.138		2.262		0.056		0.037
0155	2.250	57.1	2.347		0.049		0.138		2.324		0.056		0.037

* = Tolerance

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TABLE I. Medium duty internal retaining rings - Continued.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH D_1
	A		G	Tol	F	Tol	E	Tol	C	Tol	D	Tol	
0156	2.312	58.7	2.413		0.049		0.138		2.390		0.056		0.039
0157	2.375	60.3	2.476		0.049		0.138		2.453		0.056		0.039
0158	2.437	61.9	2.543		0.049		0.148		2.519		0.056		0.041
0159	2.440	62.0	2.546		0.049		0.148		2.522		0.056		0.041
0160	2.500	63.5	2.606		0.049		0.148		2.582		0.056		0.041
0161	2.531	64.2	2.641		0.049		0.148		2.617		0.056		0.043
0162	2.562	65.1	2.673		0.049		0.148		2.648		0.056		0.043
0163	2.625	66.6	2.736		0.049		0.148		2.711		0.056	+0.004 -0.000	0.043
0164	2.677	68.0	2.789	+0.025 -0.000	0.049		0.158		2.767		0.056		0.045
0165	2.688	68.2	2.803		0.049		0.158		2.778		0.056		0.045
0166	2.750	69.8	2.865		0.049		0.158		2.841		0.056		0.045
0167	2.813	71.4	2.929		0.049		0.158		2.903		0.056		0.045
0168	2.834	71.9	2.954		0.049		0.168		2.928		0.056		0.047
0169	2.875	73.0	2.995		0.049		0.168		2.969		0.056		0.047
0170	2.937	74.5	3.058		0.049		0.168		3.031		0.056		0.047
0171	2.952	75.0	3.073		0.049		0.168		3.046		0.056		0.047
0172	3.000	76.1	3.122		0.061		0.168		3.096		0.068		0.048
0173	3.062	77.7	3.186		0.061		0.168		3.158		0.068		0.048
0174	3.125	79.3	3.251		0.061		0.178		3.223		0.068		0.048
0175	3.149	80.0	3.276		0.061		0.178		3.247		0.068		0.048
0176	3.187	81.0	3.311		0.061		0.178		3.283		0.068		0.048
0177	3.250	82.5	3.379		0.061		0.178		3.350		0.068		0.050
0178	3.312	84.4	3.446		0.061		0.188		3.416		0.068		0.052
0179	3.346	84.9	3.479		0.061		0.188		3.450		0.068		0.052
0180	3.375	85.7	3.509		0.061	±0.003	0.188		3.479		0.068		0.052
0181	3.437	87.2	3.574		0.061		0.188		3.543		0.068		0.053
0182	3.500	88.8	3.636		0.061		0.188		3.606		0.068		0.053
0183	3.543	89.9	3.684	+0.030 -0.000	0.061		0.198	+0.004 -0.006	3.653		0.068		0.055
0184	3.562	90.5	3.703		0.061		0.198		3.672	±0.006	0.068		0.055
0185	3.625	92.0	3.769		0.061		0.198		3.737		0.068		0.056
0186	3.687	93.6	3.832		0.061		0.198		3.799		0.068		0.056
0187	3.740	95.0	3.885		0.061		0.198		3.852		0.068		0.056
0188	3.750	95.2	3.894		0.061		0.198		3.862		0.068		0.056
0189	3.812	96.8	3.963		0.061		0.208		3.930		0.068		0.059
0190	3.875	98.3	4.025		0.061		0.208		3.993		0.068		0.059
0191	3.938	99.9	4.089		0.061		0.208		4.056		0.068	+0.005 -0.000	0.059
0192	4.000	101.5	4.157		0.061		0.218		4.124		0.068		0.062
0193	4.063	103.2	4.222		0.061		0.218		4.187		0.068		0.062
0194	4.125	104.8	4.284		0.061		0.218		4.249		0.068		0.062
0195	4.188	106.5	4.347		0.061		0.218		4.311		0.068		0.062
0196	4.250	107.9	4.416		0.061		0.228		4.380		0.068		0.065
0197	4.312	109.7	4.479		0.061		0.228		4.442		0.068		0.065
0198	4.330	109.9	4.497		0.061		0.228		4.460		0.068		0.065
0199	4.375	111.0	4.543		0.061		0.228		4.505		0.068		0.065
0200	4.437	112.7	4.611		0.061		0.238		4.573		0.068		0.069
0201	4.500	114.2	4.674		0.061		0.238		4.636		0.068		0.069
0202	4.527	115.0	4.701		0.061		0.238		4.663		0.068		0.069
0203	4.562	116.0	4.737	+0.035 -0.000	0.061		0.238		4.698		0.068		0.069
0204	4.625	117.4	4.803		0.072		0.250		4.765		0.079		0.070
0205	4.687	119.0	4.867		0.072		0.250		4.827		0.079		0.070
0206	4.724	120.0	4.903		0.072		0.250		4.864		0.079		0.070
0207	4.750	120.6	4.930		0.072		0.250		4.890		0.079		0.070
0208	4.812	122.2	4.993		0.072	±0.004	0.250		4.952		0.079		0.070
0209	4.875	123.8	5.055		0.072		0.250		5.015		0.079		0.070
0210	4.921	125.0	5.102		0.072		0.250		5.061		0.079		0.070
0211	4.937	125.4	5.122		0.072		0.250		5.081		0.079		0.072

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TABLE I. Medium duty internal retaining rings - Continued.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH D_1
	A		G	Tol	F	Tol	E	Tol	C	Tol	D	Tol	
0212	5.000	126.9	5.185		0.072		0.250		5.144		0.079		0.072
0213	5.118	130.0	5.304	+0.035 -0.000	0.072		0.250		5.262	±0.006	0.079		0.072
0214	5.125	130.1	5.311		0.072		0.250		5.269		0.079		0.072
0215	5.250	133.2	5.436		0.072		0.250		5.393		0.079		0.072
0216	5.375	136.5	5.566		0.072		0.250		5.522		0.079		0.074
0217	5.500	139.7	5.693		0.072		0.250		5.647		0.079		0.074
0218	5.511	140.0	5.703		0.072		0.250	+0.004 -0.006	5.658		0.079	+0.005 -0.000	0.074
0219	5.625	142.8	5.818		0.072		0.250		5.772		0.079		0.074
0220	5.708	145.0	5.909	+0.045 -0.000	0.072		0.250		5.861	±0.007	0.079		0.077
0221	5.750	145.9	5.950		0.072		0.250		5.903		0.079		0.077
0222	5.875	149.2	6.077		0.072		0.250		6.028		0.079		0.077
0223	5.905	150.0	6.106		0.072		0.250		6.058		0.079		0.077
0224	6.000	152.3	6.202		0.072		0.250		6.153		0.079		0.077
0225	6.125	155.6	6.349		0.086		0.312		6.297		0.094		0.086
0226	6.250	158.8	6.474		0.086		0.312		6.422		0.094		0.086
0227	6.299	160.0	6.524		0.086		0.312		6.471		0.094		0.086
0228	6.375	162.0	6.601		0.086		0.312		6.547		0.094		0.086
0229	6.500	165.1	6.726		0.086		0.312		6.672		0.094		0.086
0230	6.625	168.3	6.863		0.086		0.312		6.807		0.094		0.091
0231	6.692	170.0	6.931	+0.055 -0.000	0.086		0.312		6.874		0.094		0.091
0232	6.750	171.4	6.987		0.086		0.312		6.932		0.094		0.091
0233	6.875	174.6	7.114		0.086		0.312		7.057		0.094		0.091
0234	7.000	177.8	7.239		0.086		0.312		7.182		0.094		0.091
0235	7.086	180.0	7.337		0.086		0.312		7.278		0.094		0.096
0236	7.125	181.0	7.376		0.086		0.312		7.317		0.094		0.096
0237	7.250	184.1	7.501		0.086	±0.004	0.312		7.442		0.094		0.096
0238	7.375	187.3	7.628		0.086		0.312		7.567		0.094		0.096
0239	7.480	190.0	7.734		0.086		0.312		7.672		0.094		0.096
0240	7.500	190.5	7.754		0.086		0.312		7.692		0.094		0.096
0241	7.625	193.7	7.890		0.086		0.312		7.827		0.094		0.101
0242	7.750	196.8	8.014		0.086		0.312		7.952		0.094		0.101
0243	7.875	200.0	8.131		0.086		0.312	+0.004 -0.008	8.077	±0.008	0.094	+0.006 -0.000	0.101
0244	8.000	203.2	8.266		0.086		0.312		8.202		0.094		0.101
0245	8.250	209.5	8.528		0.086		0.375		8.462		0.094		0.106
0246	8.267	210.0	8.546		0.086		0.375		8.479		0.094		0.106
0247	8.464	215.0	8.744		0.086		0.375		8.676		0.094		0.106
0248	8.500	216.0	8.780		0.086		0.375		8.712		0.094		0.106
0249	8.750	222.2	9.041	+0.065 -0.000	0.086		0.375		8.972		0.094		0.111
0250	8.858	225.0	9.151		0.086		0.375		9.080		0.094		0.111
0251	9.000	228.6	9.293		0.086		0.375		9.222		0.094		0.111
0252	9.055	230.0	9.359		0.086		0.375		9.287		0.094		0.116
0253	9.250	235.0	9.555		0.086		0.375		9.482		0.094		0.116
0254	9.448	240.0	9.755		0.086		0.375		9.680		0.094		0.116
0255	9.500	241.3	9.806		0.086		0.375		9.732		0.094		0.116
0256	9.750	247.8	10.068		0.086		0.375		9.992		0.094		0.121
0257	10.000	254.0	10.320		0.086		0.375		10.242		0.094		0.121
0258	10.250	260.4	10.582		0.086		0.375		10.502		0.094		0.126
0259	10.500	265.0	10.834		0.086		0.375		10.752		0.094		0.126
0260	10.750	273.0	11.095		0.086		0.375		11.012		0.094		0.131
0261	11.000	279.4	11.347		0.086		0.375		11.262		0.094		0.131

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Army – AR
Navy – OS
Air Force – 71
DLA – GS6

Preparing Activity:
DLA – GS6

(Project 5325-2022-012)

Review Activities:
Army – AT, AV, MI

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